

Mini-Briefing

Pattern Matching and Machine Learning for Audio Signal Processing

MA-INF 2212

#3 Fourier Transforms

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- In this lecture: four versions of Fourier Transforms (FT)
- Signal transforms for different domains (signal space FT is defined on)
 - Continuous vs discrete signals
 - Finite vs infinite signals
- FT maps signal from input **domain** to output **range**
- Both domain and range are signal spaces (can be different!)
- All FTs constructed using complex exponential functions

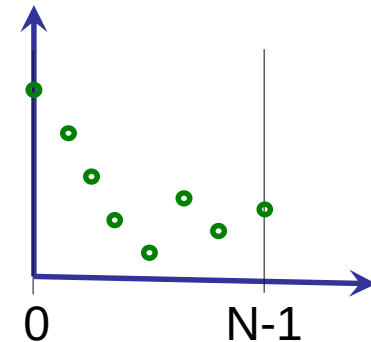
$$t \mapsto \exp(2\pi i k t)$$

- k denotes frequency (#oscillations in unit interval $[0,1)$)

Discrete Fourier Transform (DFT)

finite, discrete

- Input: $x \in \mathbb{C}^N$
- Output: $\hat{x} \equiv X := (X(0), X(1), \dots, X(N-1))^T \in \mathbb{C}^N$



- k-th Fourier coefficient obtained by DFT: (matrix * vector)

$$X(k) := \langle x | u_k \rangle = \sum_{n=0}^{N-1} x(n) \overline{\exp(2\pi i k n / N)} = \sum_{n=0}^{N-1} x(n) \exp(-2\pi i k n / N).$$

- Reconstruction/inverse transform

$$x = \frac{1}{N} \sum_{k=0}^{N-1} \langle x | u_k \rangle u_k =: \frac{1}{N} \sum_{k=0}^{N-1} X(k) u_k.$$

DFT usually
implemented
as FFT

Fourier Transform

infinite, continuous domain

- Input: $f \in L^2(\mathbb{R})$

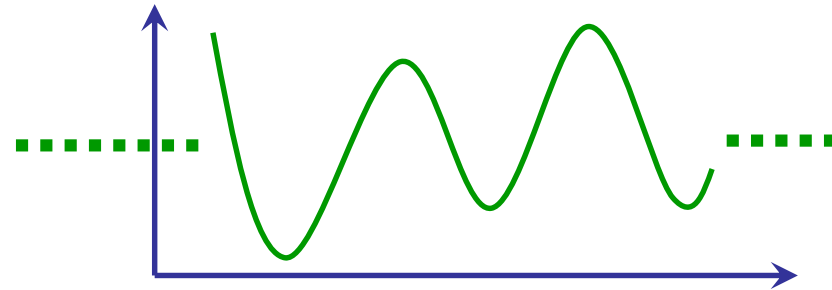
- Output: $\hat{f} \in L^2(\mathbb{R})$

- Fourier Transform

$$\hat{f}(\omega) := \int_{\mathbb{R}} f(t) e^{-2\pi i \omega t} dt.$$

- Inverse Fourier transform (essentially symmetric)

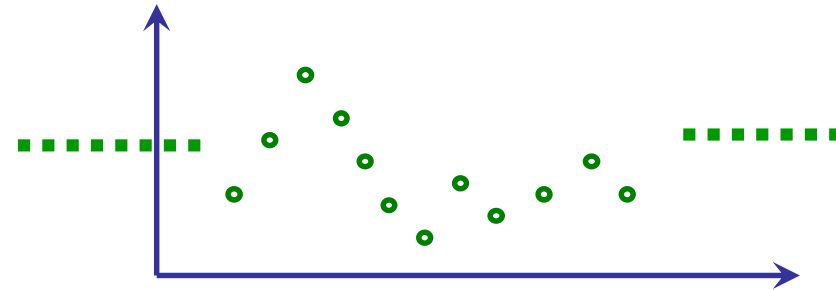
$$f(t) = \int_{\mathbb{R}} \hat{f}(\omega) e^{2\pi i \omega t} d\omega$$



Discrete time Fourier transform

infinite, discrete domain

- Input: $x \in \ell^2(\mathbb{Z})$
- Output: $\hat{x} \in L^2([0, 1])$
- Discrete time Fourier Transform



$$\hat{x}(\omega) := \sum_{k \in \mathbb{Z}} x(k) e^{-2\pi i \omega k}, \quad \text{for } \omega \in [0, 1]$$

- Fourier representation

$$x(n) = \int_0^1 \hat{x}(\omega) e^{2\pi i \omega n} d\omega$$

Fourier Series

finite, continuous domain

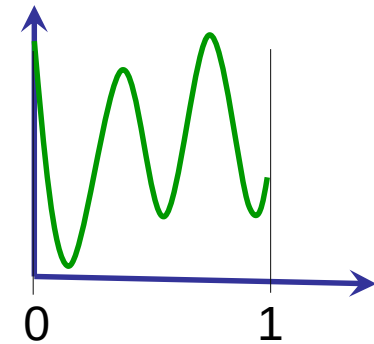
- Input: $f \in L^2([0, 1])$
- Output: $\hat{f} := (\langle f | e_k \rangle)_{k \in \mathbb{Z}} \in \ell^2(\mathbb{Z})$
- Fourier coefficients

$$\langle f | e_k \rangle = \int_0^1 f(t) e^{-2\pi i k t} dt$$

where $e_k(t) := e^{2\pi i k t}$ for $t \in [0, 1]$.

- Fourier Series representation

$$f = \sum_{k \in \mathbb{Z}} \langle f | e_k \rangle e_k.$$



Summary: Fourier Transforms

	continous domain	discrete domain
infinite domain	Fourier Transform $f \in L^2(\mathbb{R}) \longrightarrow \hat{f} \in L^2(\mathbb{R})$	Discrete time Fourier transform $f \in \ell^2(\mathbb{Z}) \longrightarrow \hat{f} \in L^2([0, 1])$
finite domain	Fourier Series $f \in L^2([0, 1]) \longrightarrow \hat{f} \in \ell^2(\mathbb{Z})$	Discrete Fourier transform (DFT) $x \in \mathbb{C}^N \longrightarrow \hat{x} \in \mathbb{C}^N$ ranges